Reg. No. | | | | | | | | | | | [17EE43]

FOURTH SEMESTER B.TECH EXAMINATION, JUNE–2019
CONTROL SYSTEMS

Time:3 Hours **Maximum Marks:70**

Instructions:

1. Answer any **FIVE** full Questions.
2. Missing data may be suitably assumed.

PART-A

1. a. Bring out the functional difference between open loop and closed loop systems and what are the effects of feedback. **04**

b. Explain the Closed Loop Control Systems with three practical applications. **04**

c. What is meant by analogous System? and obtain the dynamic equations for the mechanical system shown in fig.1c, and obtain its electrical analogue on T-V mode. **06**

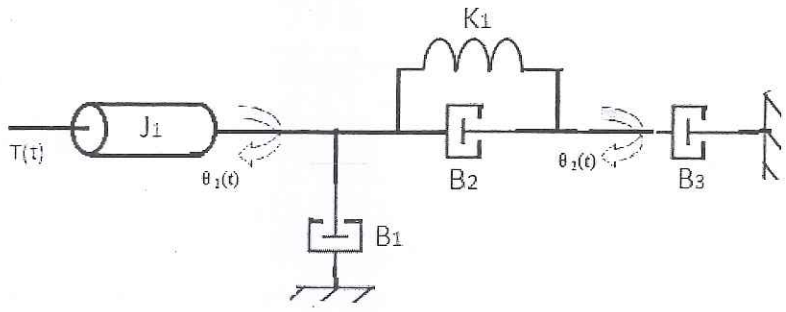

Fig.1c.

2. a. Define what is a 'block diagram'? Find the following for the system shown in the block diagram of fig 2a. **06**

- **i.** Open loop transfer function,
- **ii.** Feed forward transfer function,
- **iii.** control ratio
- **iv.** Feedback ratio
- **v.** error ratio,
- **vi.** Closed loop transfer function
- **vii.** Characteristic equation.

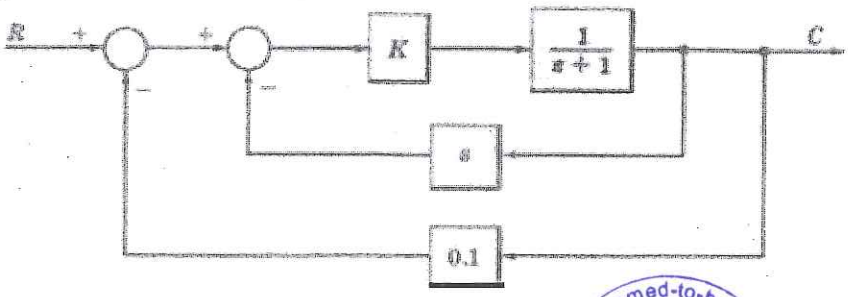

fig.2a.

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b. What are the basic properties of Signal Flow Graph? For the bridge circuit shown in fig. 2b, draw the signal flow graph and obtain the transfer function I_0/V_1 using Mason's gain formula. fig.2b. **08**

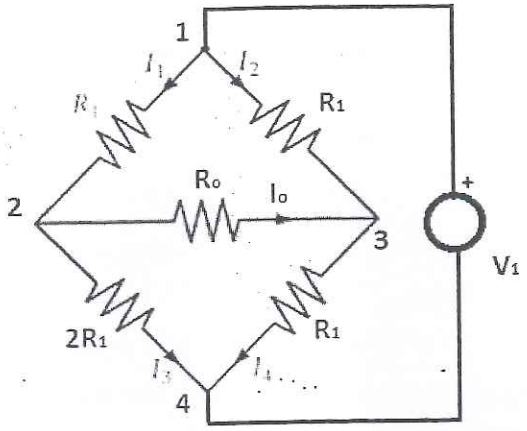


fig.2b.

3. a. What is time response of a control system? How time response of the system is divided? What is the order of the system? **03**

b. The unit step response of control system is shown in fig. 3b. Find the transfer function of the system. **05**

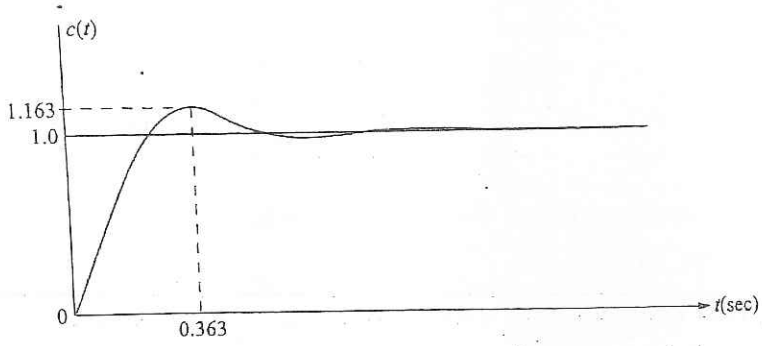


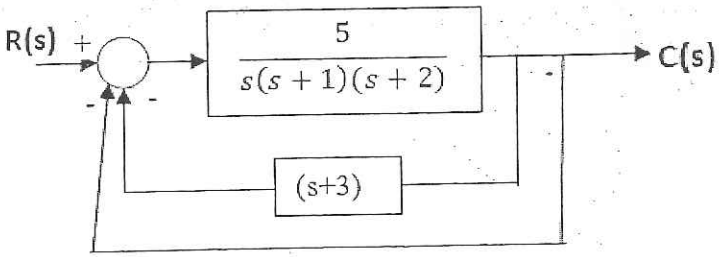
fig.3b.

c. For the control system shown in fig.3c, find the following:

i. system type,

ii. static error constants,

iii. steady state error for an input r(t) = 5u(t).



06

fig.3c.

4. a. Discuss the limitations of Routh's criterion. **03**

b. Explain bounded input and bounded output stability. **03**

c. For the block diagram shown in fig.4c, find the range of K as a function of K_1 such that the system is stable and comment on the effect of increasing the value of K_1 on the limit of gain K for stability. **08**

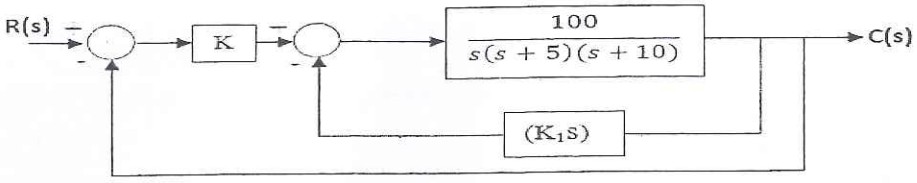


fig.4c

PART-B

5. a. What do you mean by i) breakaway point and ii) break-in point? How can they be determined? **05**

b. A block diagram representation of a unity feedback control system is shown in fig. 5b. Sketch the root locus for this system. **09**

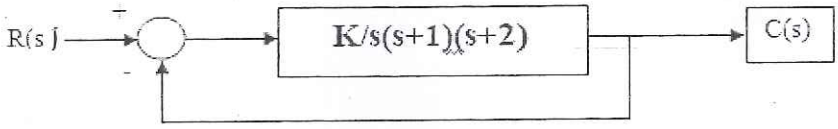


Fig.5b.

6. a. What is a polar plot and what are the advantages of it and how it is plotted? **05**

b. Sketch the Nyquist plot and there from assess the stability of the closed loop system whose open loop transfer function is $G(s)H(s) = \frac{K(s+4)}{s^2(s+1)}$ **09**

7. a. Give the correlation between time domain specifications and frequency domain specifications. **04**

b. A unity feedback control system has a transfer function of **10**

$$G(s)\,H(s) = \frac{K(s+4)}{(s-1)(s-2)}$$

i. for K=8, draw the bode plots and find there from the phase margin and gain margin.

ii. what should be the value of K for a phase margin of 30^0 and what is the corresponding gain margin?

8. a. Explain what is **i.** lead compensator and **ii.** lag compensator and what are its applications. **06**

b. The characteristic equation of an uncompensated system is given by

$$\mathbf{1 + G(s)\,H(s) = 1 + \frac{K}{s^2} = 0}$$

Whose root locus is on the jω axis. Design a compensator so that it meets the specification of settling time at 2 percent criterion of $T_s \leq 4$ seconds and damping ratio of 0.4. **08**